

TrES-3b

R-2511

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110514 · created 2026-08-25T00:18:37+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2455695.86087 +/- 0.00092 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.133 +/- 0.015
Transit depth (Rp/Rs)^2	1.76 +/- 0.39 [%]
Orbital Inclination (inc)	86.95 +/- 1.09
Airmass coefficient 1 (a1)	0.803 +/- 0.0071
Airmass coefficient 2 (a2)	0.214 +/- 0.039
Scatter in the residuals of the lightcurve fit is	0.89 %
Best Comparison Star	#3 - [510, 433]
Optimal Aperture	3.66
Optimal Annulus	13.39
Transit Duration (day)	0.0785 +/- 0.0028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.133 ± 0.015 vs 0.16309 (deviation 2.01σ)
Duration fit vs published	0.0785 d vs 0.0567 d (deviation 7.8σ)
Mid-time O–C	-3.0 min
Depth SNR	8.9
Residual scatter	0.89 %

Light curve

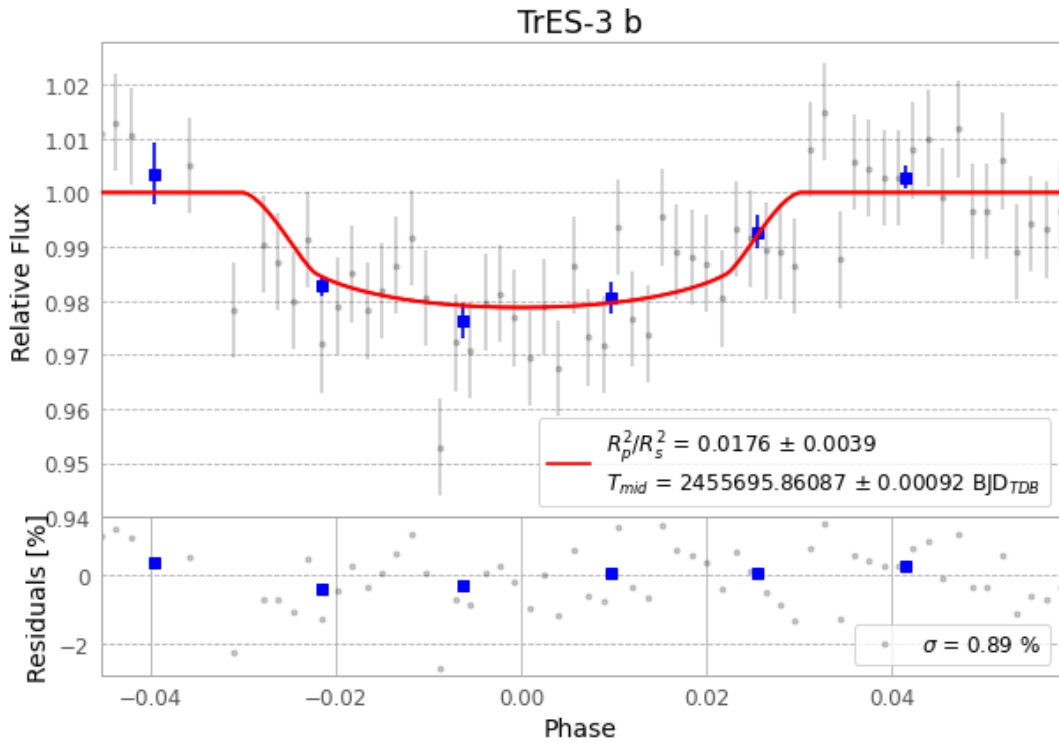


Figure 1 — FinalLightCurve_TrES-3 b_2011-05-14.png (SHA-256 2b209b4400420f5e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [273, 213], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 80f6a52f1db9ab71...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), TRES-3110514070913.FITS → TRES-3110514102413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110514.log (f660f3b4134f...), FinalLightCurve_TrES-3 b_2011-05-14.png (2b209b440042...), FinalLightCurve_TrES-3 b_2011-05-14.csv (ee641776cc75...)

Compute resources

Wall time 112.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 00:18Z.